CS-1101
PROGRAMMING FUNDAMENTALS

Lab 06 & 07

By

Engr. Muhammad Awais

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Lab Contents

- Control Structures
- Decision Control Structures
  - if - else if – else
  - Nested if

Lab Objectives

- The objective of this lab is to learn how to use **if-else if-else** ladder and **nested if** statements to control the flow of a program based on different conditions and inputs.
- Students will practice:
  - Writing **if-else if-else** ladder statements to evaluate multiple conditions in a sequence.
  - Using **nested if** statements to check conditions within other conditions, enabling more complex decision-making.
  - Understanding how to structure programs with multiple conditional branches based on various inputs.

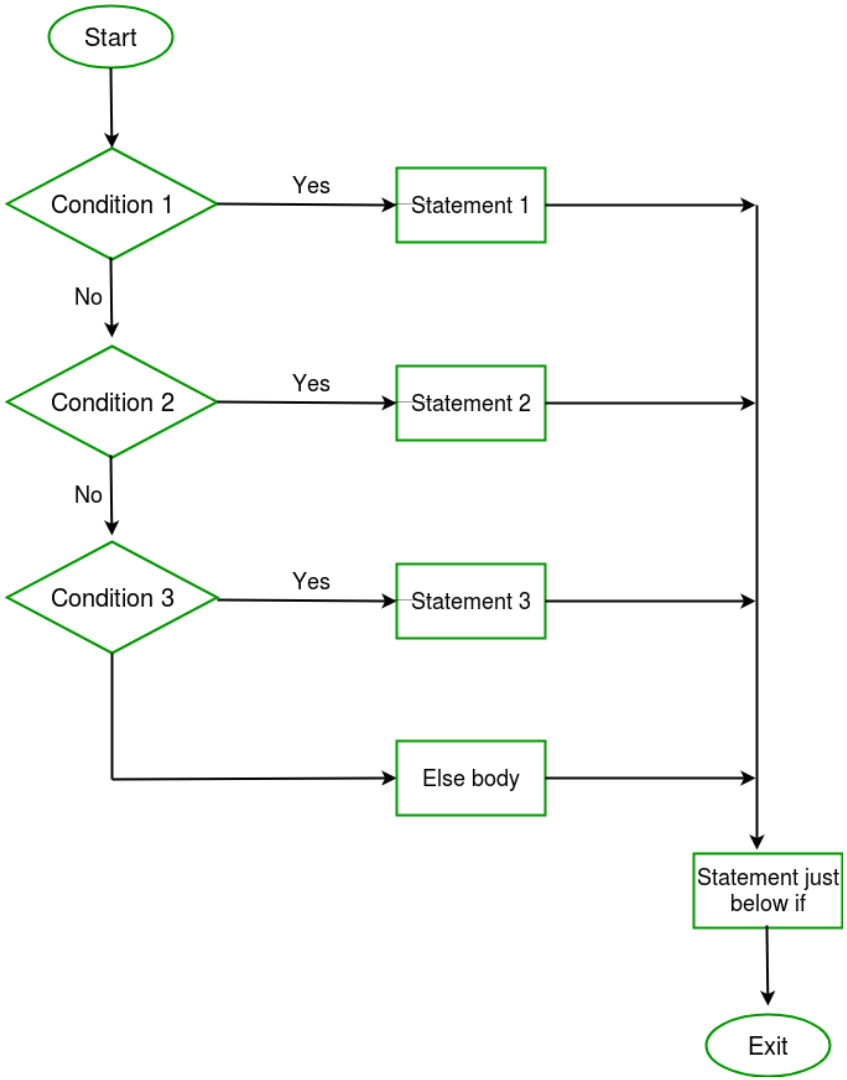
C++ if else if Ladder

- In C++, the if-else-if ladder helps the **user decide from among multiple options.**
- The C++ if statements are executed from the top down. As soon as one of the conditions controlling the if is true, the statement associated with that if is executed, and the rest of the C++ else-if ladder is bypassed.
- If none of the conditions is true, then the final statement will be executed.

Syntax

```
if (condition1) {
  // block of code to be executed if condition1 is true
} else if (condition2) {
  // block of code to be executed if the condition1 is false and condition2 is true
} else {
  // block of code to be executed if the condition1 is false and condition2 is false
}
```

C++ if else if Ladder



Issues with Multiple if Statements

- **Inefficiency:**

Even if one condition matches (e.g., grade == 'A'), the program still checks all the remaining conditions, leading to unnecessary computations.

- **Incorrect Output:**

If multiple conditions overlap, you might get conflicting outputs.

```
Enter the student's marks: 95
Grade: A
Grade: B
Grade: C
```

```
#include <iostream>
using namespace std;
int main() {
    int marks;
    cout << "Enter the student's marks: ";
    cin >> marks;
    // Overlapping conditions
    if (marks >= 90) {
        cout << "Grade: A\n";
    }
    if (marks >= 80) {
        cout << "Grade: B\n";
    }
    if (marks >= 70) {
        cout << "Grade: C\n";
    }
    if (marks < 70) {
        cout << "Grade: F\n";
    }
    return 0;
}
```

Same Program with if – else if – else Ladder

- **Efficiency:** As soon as a matching condition is found, the rest of the conditions are skipped.
- **Clear Logic:** The structure ensures that only one block is executed, avoiding conflicting outputs.

```
Enter the student's marks: 95
Grade: A
```

```
#include <iostream>
using namespace std;

int main() {
    int marks;
    cout << "Enter the student's marks: ";
    cin >> marks;
    if (marks >= 90) {
        cout << "Grade: A\n";
    } else if (marks >= 80) {
        cout << "Grade: B\n";
    } else if (marks >= 70) {
        cout << "Grade: C\n";
    } else {
        cout << "Grade: F\n";
    }
    return 0;
}
```

Nested conditions in C++

- Nested conditions are a series of conditional statements where one condition is contained within another.
- This is commonly used in programming, logic, or scenarios where multiple levels of decision-making are required.

```
// if base_condition is true control goes to base_condition1
if ( base_condition)
{
    // if base_condition is true control goes to base_condition2
    if(base_condition1)
    {
        if(base_condition2)
            ........................
        ........................
    }
}
```

Basic Tasks – if-elseif-else

- Write a program that takes an integer number as an input from user and checks if it positive , negative or zero.
- Write a program that suggests actions based on the temperature entered by the user. The program will take the temperature as input and provide advice for the user based on the range the temperature falls into. The suggestions will be:
  - If the temperature is **40°C** or higher, the program should advise the user to stay indoors because it is extremely hot and could be dangerous to health.
  - If the temperature is between **30°C and 39°C**, the program should suggest drinking plenty of water to stay hydrated in the hot weather.
  - If the temperature is between **20°C and 29°C**, the program should inform the user that the weather is pleasant and suitable for outdoor activities.
  - If the temperature is below **20°C**, the program should recommend wearing warm clothes to stay comfortable in the cooler weather.

Basic Tasks – if-elseif-else

- Write a program to simulate a simple login system. The program will prompt the user to enter a username and password. It will then compare the input values with pre-defined, correct credentials to determine the appropriate response.

Predefined Correct Credentials:

- **Username (string):** admin
- **Password (string):** 12345

The program will provide feedback based on the following scenarios:

- Both username and password are correct: The program will display "**Access Granted.“**
- Username is correct but the password is incorrect: The program will display "**Wrong Password**.“
- Username is incorrect (regardless of the password): The program will display "**User Not Found."**

University Exam Results and Scholarship Eligibility System

- You are tasked with creating a program for a university to analyze and classify students' results based on their performance in three subjects. The program must calculate the total marks, percentage, grade, and eligibility for a scholarship based on specific conditions.

Conditions:

Grade Calculation:

- Percentage >= 90: Grade A
- Percentage >= 80 and < 90: Grade B
- Percentage >= 70 and < 80: Grade C
- Percentage >= 60 and < 70: Grade D
- Percentage < 60: Grade F

Failure Condition:

A student is considered failed if:

- They score Grade F in any subject.

Scholarship Eligibility:

A student is eligible for a merit scholarship if:

- They score Grade A in all three subjects.
- The total marks are >= 270.

A student is eligible for a regular scholarship if:

- They score at least Grade B in all three subjects.
- The total marks are >= 240.

Output Details:

- Display the total marks and percentage.
- Show the grade for each subject.
- Indicate if the student is eligible for a scholarship (and which type).
- If the student has failed, display "Failed" and terminate further calculations

Loan Eligibility Checker --- Nested if

You work as a software developer for a bank. The bank needs a program that checks whether a person qualifies for a loan based on two criteria: age and income.

The bank's policies for loan eligibility are as follows:

Age Requirement: The person must be at least 18 years old to be considered for a loan. If they are younger than 18, the program should display a message:

"*You are not eligible for a loan because you are under 18 years of age*."

Income Requirement: The person must have an annual income of at least $30,000.If they meet the age requirement but their income is less than $30,000, the program should display a message:

"You meet the age requirement but are not eligible for a loan because your income is below $30,000."

Eligible Applicants: If the person meets both criteria, the program should congratulate them and display*:*

"Congratulations! You are eligible for a loan."

Your task is to write a program that takes the age and income of a person as input and checks their eligibility based on the above rules. The program should give clear and descriptive messages for all possible cases.

Role Based Access System

- You are developing an Access Control System for a software platform. The system will first authenticate the user based on a predefined username and password. After successful authentication, the system will check the user's role to determine the level of access they should have. The roles can be either "Admin," "Guest," or any other custom role. The access permissions are as follows:

Authentication:

The system will compare the entered username and password with the predefined credentials.

- If the username is "admin" and the password is "password123" the authentication is successful. The system will display:

  "Authentication Successful!"

- If the username or password does not match the predefined values, the authentication will fail. The system will display:

  "Authentication Failed. Access Denied."

Role-Based Access:

- If the role is "Admin": The system will display: "Full Access"
- If the role is "Guest": The system will display: "Limited Access"
- For any other role (e.g., "User," "Manager"): The system will display: "No Access"

REFERENCES

- https://www.geeksforgeeks.org/c-c-if-else-if-ladder-with-examples/